

Internship Report: Web Application Security Assessment

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1. Executive Summary

The primary objective of this internship was to gain hands-on experience in web application security, specifically focusing on identifying and mitigating common vulnerabilities. This report details the methodology, tools utilized, and the successful identification of security flaws through both automated scanning and manual exploitation techniques.

2. Tools & Environment

- **Automated Security Tool:** OWASP ZAP (Version 2.17.0)
- **Training Platform:** PortSwigger Web Security Academy
- **Target Environment:** testasp.vulnweb.com and PortSwigger Lab environments

3. Vulnerability Assessment & Methodology

3.1 Automated Scanning

Using OWASP ZAP, a comprehensive vulnerability scan was performed on the target web application. The scan identified 8 vulnerabilities, including high-risk SQL Injection and medium-risk Cross-Site Scripting (XSS), which could allow unauthorized database access or malicious script execution.

SQL Injection

URL: <http://testasp.vulnweb.com/Search.asp?tfSearch=ZAP%27>

Risk:  High

Confidence: Low

Parameter: tfSearch

Attack: ZAP

Evidence: HTTP/1.1 500 Internal Server Error

CWE ID: 89

WASC ID: 19

Source: Active (40018 - SQL Injection)

Input Vector: URL Query String

Description:

SQL injection may be possible.

Other Info:

Solution:

Do not trust client side input, even if there is client side validation in place.
In general, type check all data on the server side.
If the application uses JDBC, use PreparedStatement or CallableStatement, with parameters passed by '?'

Reference:

https://cheatsheetseries.owasp.org/cheatsheets/SQL_Injection_Prevention_Cheat_Sheet.html

Alert Tags:

Key	Value
POLICY_PENTEST	
HIPAA	https://www.zaproxy.org/docs/desktop/addons/common-library/alerttags/#compliance
OWASP_2017_A01	https://owasp.org/www-project-top-ten/2017/A1_2017-Injection.html
API_2023_API10	https://owasp.org/API-Security/editions/2023/en/0xaa-unsafe-consumption-of-apis/
POLICY_DEV_STD	

5. Conclusion

This internship provided critical insights into the lifecycle of web vulnerabilities. Through practical application, I have demonstrated the ability to detect security weaknesses and propose effective remediation strategies, establishing a solid foundation for future professional engagements in cybersecurity.